FIN Local Research — Release 10.45

Checkpoint P474–P479–P483

Complex Optimal-Face Evidence and an Exact Parametric O181 Tube

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Exact result. A uniform rational Krawczyk inclusion proves a unique positive O181 polynomial root in one common box for every

$$|q - 4/5| \leq 3 \times 10^{-9}, \qquad |\theta - \pi/8| \leq 3 \times 10^{-9}.$$

The corresponding O167 point attains the global value of the declared reduced three-slot ordered-comb SDP throughout this rectangle.

Optimal-face result. Two independent finite linearizations show one candidate complex flat direction through the real optimum. This is strong numerical evidence, not yet an exact nonuniqueness theorem.

Boundary. No selector, dimensional source, laboratory record, complete legacy–strict bridge, role transfer, or physical closure is claimed. The exported Lean source was not machine checked because no local toolchain is installed.

Publication statement

This monograph reports local analytical and computer-assisted Programs P474, P479, and P483. It uses no laboratory record, external audit, internet result, remote computation, or physical validation.

Contents

Chapter 1

Executive summary

This checkpoint executes local Programs P474, P479, and P483 after the exact O167 attainment theorem of P473. The programs use only analytical derivation, finite linear algebra, exact rational interval arithmetic, and local code. No internet source, external audit, remote model, laboratory record, or empirical physical evidence is used.

The principal results are:

1. **P474 — [Strong evidence].** The exact O167 root is locally unique in its 13-dimensional real polynomial chart, but the complete complex causal comb appears not to have a unique optimizer. The full linearized complementarity-and-causality system has size 299×128, numerical rank (127), and one null direction. Its smallest nonzero singular value is $1.8591352597 \times 10^{-2}$, while the remaining singular value is 6.96×10^{-15}. An independent 28×7 imaginary-causal Riccati tangent map has rank six and the same one-dimensional defect. Along the tested segment $t \in [-0.025, 0.025]$, the normalizer remains positive, with minimum sampled eigenvalue 9.61×10^{-3}, while the objective changes by at most 1.12×10^{-15}. This is compelling evidence for a one-dimensional complex optimal face, but an exact algebraic null-vector certificate is still missing. Full-cone uniqueness is therefore not marked refuted.
2. **P479 — [Open as machine formalization].** A dependency-free Lean source now separates the Riccati-to-support implication, trace-telescope contact, weak duality, and the distinction between local root uniqueness and global optimizer uniqueness. The local lean launcher exists, but no Lean toolchain is installed. In accordance with the local-only rule, no network installation was attempted. The file is an explicit formal interface, not a machine-checked theorem.
3. **P483 — [Computer-assisted proof].** The O181 root is not an isolated artifact of the single point $q = 4/5, \theta = \pi/8$. A uniform exact rational Krawczyk calculation proves that, for every

$$|q - 4/5| \leq 3 \times 10^{-9}, \qquad |\theta - \pi/8| \leq 3 \times 10^{-9},$$

   there exists a unique positive root in one common 13-dimensional box of radius 10^{-5}. The exact inclusion margin is greater than 4.7195×10^{-6}, and the maximum interval contraction row sum is less than (0.229493). Uniform positivity bounds are

$$N \succeq \frac{6}{625} I, \qquad X_3 \succeq \frac{31}{3125} I.$$

   The same Riccati, block-equality, and trace-telescope argument as P473 then proves exact primal–dual attainment throughout this parameter rectangle. Root uniqueness is local to the common polynomial box; uniqueness of the optimizer over the complete complex cone is not claimed.

The new objects are O182 Complex Comb Optimal-Face Direction, O183 Riccati-Attainment Formal Interface, and O184 Parametric O181 Krawczyk Tube.

No FIN selector, QW-2191 discharge, dimensional source, physical clock, preparation, apparatus, independent measurement record, complete legacy-to-strict bridge, legacy role transfer, L_{total}, Standard Model or gravitational closure, or Theory of Everything is claimed.

Chapter 2

Confidence convention

Label	Meaning
[Proven]	Exact analytic or finite symbolic consequence of the declared premises
[Computer-assisted proof]	Finite rational or outward-interval certificate with auditable local artifacts
[Strong evidence]	Reproducible numerical evidence lacking one exact certificate
[Conditional]	Conclusion dependent on an explicitly named mathematical interface
[Open]	Current local artifacts do not decide the proposition
[Refuted]	A candidate fails a necessary test or an exact counterexample exists
[Blocked by external evidence]	Laboratory, calibration, custody, or independent external evidence is indispensable

Chapter 3

State transition and scientific boundaries

Lane	Release 10.44 state	Release 10.45 state
Exact O167 attainment at $q = 4/5, \theta = \pi/8$	Proved	Preserved
Local polynomial-root uniqueness	One certified point box	Uniform common box over a certified parameter rectangle
Parameter robustness	Open	Proved for radii 3×10^{-9} in q and θ
Full complex optimal face	Open	One-dimensional face strongly indicated; exact certificate open
Lean formalization	Not attempted in P471–P473	Source exported; local checker unavailable
Selector / QW-2191	Open	Unchanged
Dimensional source	Missing	Unchanged
Laboratory evidence	None	None
Legacy-to-strict bridge	Incomplete	Unchanged

The kernel split remains binding:

$$K_{\mathrm{legacy,ont}}(d) = \frac{\alpha_{\mathrm{geo}} \cos(\omega d + \phi)}{1 + \beta_{\mathrm{tors}} d}, \qquad K_{\mathrm{strict,gate}}(d) = \frac{\cos(\omega d + \phi)}{1 + \beta d^{\eta}}.$$

The current programs are downstream analyses of supplied finite channel operators. They derive neither kernel, complete no bridge, and transfer no legacy physical role. The ontology remains

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer},$$

with no informational substrate below the nadsoliton.

Chapter 4

Inherited exact problem

4.1 Ordered three-slot primal

For the supplied compressed Hermitian process difference Δ, the reduced ordered three-slot primal is

$$\begin{array}{rl} \text{maximize} & \frac{1}{2}\operatorname{Tr}[\Delta(T_+ - T_-)] \\ \text{subject to} & T_+ + T_- = B_0 \oplus B_1, \\ & B_0 + B_1 = C_0 \oplus C_1, \\ & C_0 + C_1 = \operatorname{diag}(d_0, d_1), \\ & d_0 + d_1 = 1, \end{array}$$

where all primal matrices are positive semidefinite. For a fixed positive normalizer $N = T_+ + T_-$, the inner binary discrimination problem is

$$F(N) = \frac{1}{2}\left\| N^{1/2}\Delta N^{1/2} \right\|_1 .$$

4.2 Nested dual and P473 contact

The dual minimizes λ under

$$X_3 \succeq \pm\frac{\Delta}{2},$$

followed by the nested block majorization ladder

$$X_2 \succeq (X_3)_{00}, (X_3)_{11}, \qquad X_1 \succeq (X_2)_{00}, (X_2)_{11}, \qquad \lambda \geq (X_1)_{00}, (X_1)_{11}.$$

P473 proved one exact positive root of

$$X_3 N X_3 = \frac{1}{4}\Delta N \Delta$$

inside a rational box of radius 3×10^{-14}. The structured form of X_3 makes all lower dual slacks vanish, while the positive Riccati theorem gives the top support and the trace telescope gives

$$F(N) = \operatorname{Tr}(N X_3) = \lambda.$$

Thus one exact global optimizer was known before P474. What remained open was whether every global optimizer must coincide with that one.

Chapter 5

P474 — complete optimal-face audit

5.1 Why local root uniqueness is insufficient

The 13 P471 variables describe a real symmetry chart for N and X_3. Krawczyk uniqueness in this chart means only that no second root occurs inside that local real box. The primal cone, however, contains complex Hermitian normalizers and complex Hermitian tester effects. A global uniqueness theorem must exclude those directions as well.

At an exact dual optimum define the top slacks

$$S_+ = X_3 - \frac{\Delta}{2}, \qquad S_- = X_3 + \frac{\Delta}{2}.$$

If two primal optima differ by $\delta T_+, \delta T_-$, complementary slackness and homogeneous causality require

$$S_+ \delta T_+ = 0, \qquad S_- \delta T_- = 0,$$

together with the homogeneous version of every nested comb constraint for $\delta N = \delta T_+ + \delta T_-$.

Each 8×8 Hermitian matrix has 64 real coordinates. The two tester directions therefore have 128 unknowns. Expanding both complex contact equations and the 43 independent real causal equations produces a 299×128 real matrix.

5.2 Rank defect

At the P473 center the singular-value calculation gives

$$\operatorname{rank}_{10^{-10}} M_{\text{contact}} = 127.$$

The two terminal singular values are

$$\sigma_{127} = 1.859135259729509 \times 10^{-2}, \qquad \sigma_{128} = 6.962781439294319 \times 10^{-15}.$$

The separation of more than twelve orders of magnitude is incompatible with an ordinary poorly conditioned full-rank matrix. It identifies one candidate flat direction and leaves all transverse directions well separated from zero.

5.3 Independent Riccati tangent representation

The channel difference is purely imaginary. Write

$$\Delta = iK, \qquad K^T = -K,$$

with K real. A complex Hermitian normalizer perturbation can be written

$$N(t) = N_* + itQ, \qquad Q^T = -Q,$$

where Q is a real skew matrix. Because the Riccati equation is linear in N, the whole affine line satisfies it exactly if

$$\boxed{X_3QX_3 + \frac{1}{4}KQK = 0.}$$

The imaginary tangent space of the three-slot causal recursion has seven real coordinates: one inherited C_0 skew coordinate and six B_0 skew coordinates. The resulting Riccati tangent map has size 28×7. Its singular values are

$$(0.468366,\ 0.269691,\ 0.195477,\ 0.131330,\\ 0.0538862,\ 0.0160806,\ 2.77928 \times 10^{-15}).$$

It independently has rank six. After normalization, the null vector is numerically

$$(0, 1, 1, -\rho, 0, 1, 1), \qquad \rho = 0.0297904414933592,$$

up to errors of order 3.2×10^{-13} in the coordinates expected to be zero or one. In the upper 4×4 block this is

$$Q_0 = \begin{pmatrix} 0 & 1 & 1 & -\rho \\ -1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ \rho & -1 & -1 & 0 \end{pmatrix}, \qquad Q = Q_0 \oplus (-Q_0).$$

This form satisfies the homogeneous causal recursion by construction. The remaining task is to prove that its exact ρ satisfies every exact Riccati tangent equation at the exact P473 root.

5.4 Positivity and constant-value scout

The full tester null vector was reconstructed independently from the 299×128 contact system. Twenty-one equally spaced points were tested for

$$-0.025 \le t \le 0.025.$$

Across this segment:

- the smallest normalizer eigenvalue is at least $9.6097438913 \times 10^{-3}$;
- the smallest tester eigenvalues are zero within numerical error (-3.58×10^{-16});
- the maximum change in the objective is 1.11×10^{-15}.

The figure below shows that the candidate line remains well inside the positive normalizer cone over the tested range.

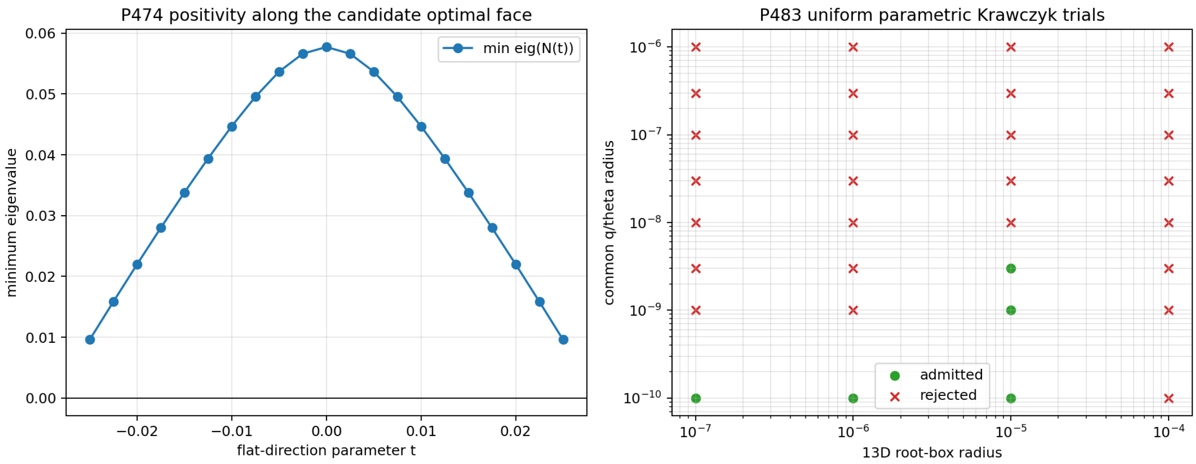

Figure 5.1: Positivity along the candidate P474 complex optimal direction and the exactly admitted/rejected P483 uniform Krawczyk tubes.

5.5 P474 verdict

[Strong evidence] The real O167 solution is a point on a one-dimensional complex optimal face, rather than the unique full-cone optimizer.

[Open] This is not yet an exact nonuniqueness theorem. The missing object is a rational/algebraic certificate that the exact P473 root makes the 28×7 tangent map singular with exact null vector $Q \neq 0$. Until that certificate exists, neither "unique" nor "nonunique" is asserted as a theorem for the complete complex cone.

5.6 Interpretation boundary

The candidate direction is an operational phase degree of freedom inside a finite complex tester cone. It is not a FIN selector, a physical time, a measured phase, a gauge field, or evidence of an observer mechanism. Such interpretations would require additional definitions and records absent from the present local mathematics.

Chapter 6

P479 — formal dependency interface

6.1 What was encoded

The file FIN_Programs_474_479_483_P479_Riccati_Trace_Core.lean contains four dependency-free theorem interfaces:

1. riccati_support_bridge isolates the analytic theorem converting a positive Riccati solution into the trace-norm support;
2. trace_telescope_attainment isolates equality produced by primal–dual contact;
3. exact_attainment_blocks_strict_improvement proves the standard weak duality consequence that no feasible primal point can exceed an attained dual value;
4. local_root_uniqueness_does_not_assert_global_optimizer_uniqueness prevents the precise overclaim investigated by P474.

The abstraction is intentional. It does not pretend that spectral square roots, matrix absolute values, positivity, and trace theory have been formalized merely because their implication structure has been typed.

6.2 Local checker result

The local executable /usr/bin/lean is only an elan launcher. It reports that no default toolchain is configured. The project constraints prohibit network access, so no toolchain was downloaded.

[Open] The source is not machine-checked in the present environment.

[Proven outside Lean] The finite-dimensional O181 result remains proved by the exact polynomial, interval, positivity, and primal–dual artifacts of P471–P473. P479 changes no confidence label attached to that result.

Chapter 7

P483 — exact parametric Krawczyk tube

7.1 Parametric polynomial family

P471 fixed $q = 4/5$ and represented the three phase quantities by

$$s_k = \sin(k\theta), \qquad k = 1, 2, 3.$$

P483 promotes q to an interval parameter and evaluates all coefficients of the 13 Riccati orbit polynomials over boxes for

$$(q, s_1, s_2, s_3).$$

For $\theta_0 = \pi/8$, the mean-value theorem supplies the exact enclosure

$$|\sin(k\theta) - \sin(k\theta_0)| \leq k|\theta - \theta_0|.$$

The calculation therefore over-approximates the true one-dimensional trigonometric curve by an independent four-parameter box. A successful certificate on this larger box is valid on the physical parameter curve.

7.2 Uniform Krawczyk map

Let (c) be the rational center inherited from P473, (B_r(c)) the 13-dimensional coefficient box, and (R) a rational approximate inverse of the nominal Jacobian. For the interval parameter box (P), P483 evaluates

$$\mathcal{K}(B_r, P) = c - RF(c, P) + \big(I - RJ(B_r, P)\big)(B_r - c)$$

using exact Fraction arithmetic. Uniform strict inclusion

$$\mathcal{K}(B_r, P) \subset \operatorname{int} B_r(c)$$

proves existence and uniqueness of a root inside (B_r(c)) for every parameter point in (P).

Thirty-two candidate tubes were tested. Five were admitted. The selected largest common parameter radius uses

$$r = 10^{-5}, \qquad \delta_q = \delta_\theta = 3 \times 10^{-9}.$$

Its rational q-interval is

$$\frac{799999997}{10^9} \leq q \leq \frac{800000003}{10^9}.$$

The exact minimum Krawczyk inclusion margin is approximately

$$4.7195001774 \times 10^{-6} > 0,$$

and the maximum row-sum bound for (I-RJ) is

$$0.2294927355 < 1.$$

The maximum interval residual at the common center is

$$1.0152457599 \times 10^{-8}.$$

7.3 Uniform positivity

The structured normalizer changes by at most (40r) in operator norm under the declared 13-dimensional box, and X_3 changes by at most (8r). Paying these bounds against the exact P473 center margins gives

$$\lambda_{\min}(N) \geq \frac{1}{100} - 40 \times 10^{-5} = \frac{6}{625} > 0,$$

and

$$\lambda_{\min}(X_3) \geq \frac{1}{100} - 8 \times 10^{-5} = \frac{31}{3125} > 0.$$

Thus every certified root in the tube remains in the positive Riccati branch.

7.4 Uniform attainment theorem

Theorem P483 — [Computer-assisted proof]. For every real pair

$$(q, \theta) \in \left[\frac{4}{5} - 3 \times 10^{-9}, \frac{4}{5} + 3 \times 10^{-9}\right] \times \left[\frac{\pi}{8} - 3 \times 10^{-9}, \frac{\pi}{8} + 3 \times 10^{-9}\right],$$

the declared 13-polynomial ordered-comb system has exactly one positive root inside the common ℓ_∞ box of radius 10^{-5} centered at the P473 rational point. For this root, the structured O167 normalizer and dual support satisfy the top Riccati support identity, all lower dual block equalities, and the trace telescope. Consequently an O167 primal point exactly attains the global value of the declared reduced three-slot SDP for every parameter pair in the rectangle.

Proof. Exact rational interval evaluation encloses all 13 residuals and 169 Jacobian entries over the joint root-and-parameter box. The rational Krawczyk image lies strictly inside the common root box, uniformly in the parameter coordinates. The explicit norm payments above retain (N>0) and $X_3 > 0$. The P471 positive Riccati theorem yields $X_3 \succeq \pm\Delta/2$. The imposed block pattern makes the lower dual slacks zero. The recursive trace identities equate the primal and dual values. Weak duality then proves global attainment.

7.5 What P483 does not prove

The theorem establishes local continuation of the exact polynomial root and global attainment of a corresponding O167 primal point. It does not:

- classify the full complex optimal face;
- prove that the same tube is maximal;
- prove behavior outside the declared q, θ rectangle;
- identify q or θ with a measured physical parameter;
- derive dimensional time, energy, length, mass, or action;
- supply an experiment, calibration, custody chain, or hold-out record.

Chapter 8

Cross-program synthesis

P473, P474, and P483 distinguish three notions that must not be conflated:

Notion	**Current result**
Unique root in a local real polynomial chart	Proved at the nominal point and uniformly in the P483 tube
Existence of a globally attaining primal point	Proved at the nominal point and throughout the P483 tube
Unique optimizer in the complete complex causal cone	Open; P474 gives strong evidence against it

This distinction resolves an apparent tension. Parametric stability of the root is compatible with degeneracy of the complete primal optimal face. The root determines one real representative and a dual certificate; it need not determine every complex tester realizing the same value.

The likely new geometry is therefore a locally stable base point carrying a one-dimensional complex optimal fiber:

$$(q,\theta) \longmapsto (N_*(q,\theta), X_3(q,\theta)) \quad \text{with a candidate optimal phase fiber over each point.}$$

This bundle language is presently an organizing intuition, not a proved topological structure. Proving the exact P474 null direction and transporting it through the P483 tube are separate obligations.

Chapter 9

Falsification ledger

Candidate claim	**Attack**	**Result**	**Status**
Local real root uniqueness implies full-cone uniqueness	Expand to all complex Hermitian tester coordinates	A one-dimensional numerical nullspace appears	[Strong evidence] against implication
Rank defect is mere ill-conditioning	Compare terminal singular values	Gap exceeds twelve orders of magnitude; next singular value > 0.0185	[Strong evidence] genuine defect
Candidate direction immediately exits positivity	Scan $t \in [-0.025, 0.025]$	Normalizer margin stays > 0.0096	[Strong evidence] robust feasible segment
Flatness is only first order	Recompute full trace-norm objective	Maximum deviation 1.11×10^{-15}	[Strong evidence] finite segment flatness
P473 is isolated at one parameter point	Uniform parameter Krawczyk test	Five strict tube certificates admitted	[Refuted]
P483 positivity could fail inside wider root box	Exact perturbation payment	Bounds $(6/625)$ and $(31/3125)$ remain positive	[Computer-assisted proof]
Lean source is already machine checked	Invoke local checker	No toolchain configured	[Refuted]
Mathematical robustness supplies physical calibration	Audit dimensions and records	No scale or record is generated	[Refuted]

Chapter 10

Reproducibility and local-resource audit

The P483 certificate uses exact rational arithmetic after symbolic generation of the polynomial system. Floating-point computations are used in P474 and to choose the rational Krawczyk preconditioner, but not to decide P483's strict inclusion inequalities.

The calculations fit the declared local hardware envelope. The expensive step is interval evaluation of 32 candidate parameter tubes; it remains a small 13×13 problem and requires neither GPU nor remote computation.

The following artifacts are included:

- fin_programs_474_479_483.py;
- test_fin_programs_474_479_483.py;
- FIN_Programs_474_479_483_Results.json;
- FIN_Programs_474_479_483_Summary.csv;
- FIN_Programs_474_479_483_P474_Optimal_Face.csv;
- FIN_Programs_474_479_483_P474_Flat_Direction.npz;
- FIN_Programs_474_479_483_P479_Formalization.csv;
- FIN_Programs_474_479_483_P479_Riccati_Trace_Core.lean;
- FIN_Programs_474_479_483_P483_Parametric_Tube.csv;
- figure directory FIN_Programs_474_479_483_Figures, containing p474_p483_optimal_face_and_parameter_tube.png.

Chapter 11

Ranked next research programs

Rank	Program	Question	Method and success criterion
1	P484	Is the P474 complex flat direction exact?	Derive a symbolic $(\mathrm{Q}\rho)$, certify one exact nonzero tangent root, and prove positivity for a nonzero rational interval in t
2	P485	Does the P474 optimal fiber persist throughout the P483 tube?	Uniform interval rank/minor certificate for the 28×7 tangent map
3	P480	Can O180/O181 be replayed without SciPy or SymPy?	Serialize monomials, boxes, rational preconditioner, and implement a standard-library-only checker
4	P475	Does the O181 value have a tractable algebraic minimal polynomial?	Elimination over the exact algebraic sine field; stop on certified resource threshold
5	P476	How large is the analytic uniqueness region behind P465?	Prove $4q^2 \cos^2 \theta > 1$ with all sector-weight boundary conditions
6	P477	Does the polynomial support ladder generalize to arbitrary slot number?	Inductive orbit count and nested Riccati theorem
7	P478	Is there a four-slot coherent optimum with an exact certificate?	Reduced four-slot primal and structured dual before any full-cone claim
8	P481	Are there other positive roots outside the P483 local box?	Certified interval branch-and-bound over a bounded feasible domain
9	P486	Is the candidate optimal fiber gauge, operational, or observable?	Quotient the tester equivalence relation and identify invariant outcome statistics without physical promotion
10	P470	Are downstream conclusions stable under strict-kernel parameter boxes?	Interval perturbation with explicit legacy/strict separation and no role transfer
11	P463	What is the minimal preparation/instrument contract?	Typed local interface for state, clock, instrument, environment, apparatus, and record; explicitly conditional
12	P460	Is allocation O174 stable under alternative admissible loss geometries?	Quotient sensitivity and certified ranking margins

The next bounded batch is

$$\boxed{P484 \longrightarrow P480 \longrightarrow P475}.$$

P484 has priority because it decides whether full-cone optimizer uniqueness is false or merely numerically doubtful. P480 protects the exact P473/P483 results against loss of large software dependencies. P475 is attempted only after those two proof-grade tasks and must stop if elimination exceeds the local resource ceiling.

Chapter 12

Conclusion

This checkpoint strengthens and sharpens the finite operator theory without promoting it to physics. P483 proves that exact O167 primal–dual attainment is stable on a nonzero, rigorously certified q, θ neighborhood. P474 shows, with two independent numerical formulations, that the corresponding real optimizer probably sits on a one-dimensional complex optimal face. P479 records the theorem dependency structure and exposes the absence of a local Lean toolchain rather than hiding it.

The deepest surviving interpretation of this batch is a robust local family of exact finite-comb discrimination certificates, possibly equipped with a complex operational degeneracy. That structure belongs to finite operator optimization and quantum-information geometry. It does not yet contain the selector, dimensional scale, operational measurement record, or empirical bridge required for a physical theory.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/fin-mpl-1045 \
python3 fin_programs_474_479_483.py
python3 -m unittest -v test_fin_programs_474_479_483.py
python3 fin_programs_474_479_483_to_latex.py
lualatex -interaction=nonstopmode -halt-on-error \
  FIN_Local_Research_Checkpoint_P474_P479_P483_EN.tex
sha256sum -c FIN_PROGRAMS_474_479_483_RELEASE_MANIFEST.sha256
```

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